

# pi515 — Iowa Soil Quality Dataset: Curation Documentation

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## Overview

This document describes the curation process for the **pi515 Iowa Soil Quality Dataset**, a paired aerial imagery and tabular soil/climate dataset built to train a multimodal machine learning model (EfficientNet-B0 with tabular fusion) to predict soil quality across Iowa corn-producing regions.

The dataset consists of approximately **1,121 labeled samples**, each comprising:

- A **224×224 pixel NAIP aerial image patch** centred on a verified Iowa corn field
- **Tabular soil metrics**: pH, organic carbon, texture fractions, bulk density
- **Seasonal climate metrics**: precipitation, temperature, vapour pressure deficit
- A **corn yield-derived quality label**: High / Medium / Low

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## Data Sources

| Source                | What It Provides                          | Resolution | Access                       |
|-----------------------|-------------------------------------------|------------|------------------------------|
| USDA NAIP (via GEE)   | RGB aerial imagery                        | ~1 m       | Free via Google Earth Engine |
| USDA CDL (via GEE)    | Corn field mask by year                   | 30 m       | Free via Google Earth Engine |
| OpenLandMap (via GEE) | pH, organic carbon, texture, bulk density | 250 m      | Free via Google Earth Engine |
| PRISM AN81m (via GEE) | Precipitation, temperature, VPD by month  | ~4 km      | Free via Google Earth Engine |
| TIGER/2016 (via GEE)  | Iowa county boundaries                    | Vector     | Free via Google Earth Engine |
| USDA NASS QuickStats  | County-level corn yield (bu/acre)         | County     | Free API                     |

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## Pipeline Architecture

The dataset was assembled using an automated 7-step Python pipeline (`pipeline.py`) built on the Google Earth Engine (GEE) Python API.

## Step 1 — GPS Point Sampling

**Goal:** Identify real Iowa corn field locations to sample from.

Sample points were drawn exclusively from pixels classified as **corn (CDL class 1)** in the USDA Cropland Data Layer for each target year. This guarantees every image patch shows an active corn-producing field, not arbitrary Iowa land.

- **Years sampled:** 2019, 2020, 2021, 2022
- **Points per year:** ~280 (750 requested per year; GEE returned available corn pixels)
- **Total points:** 1,121
- **Sampling method:** `ee.Image.sample()` with stratified random seed per year
- **State boundary:** Iowa geometry from TIGER/2018/States

## Step 2 — NAIP Aerial Image Export

**Goal:** Export a 224×224 pixel aerial image patch centred on each GPS point.

The National Agriculture Imagery Program (NAIP) provides sub-meter resolution RGB+NIR aerial imagery captured during growing season. Because NAIP is captured every 2–3 years per county, each crop year was matched to the nearest available NAIP collection year:

| Crop Year | NAIP Year Used |
|-----------|----------------|
| 2019      | 2018           |
| 2020      | 2020           |
| 2021      | 2020           |
| 2022      | 2022           |

### Patch parameters:

- Spatial extent: 50 m × 50 m centred on sample point
- Output resolution: 224 × 224 pixels (~0.22 m/pixel)
- Bands exported: R, G, B
- Output format: GeoTIFF □ Google Drive folder `pi515_iowa_soil`
- CRS: EPSG:4326

Export tasks were submitted to GEE’s asynchronous batch export system, which processed them in parallel on Google’s infrastructure.

## Step 3 — Soil Metric Extraction

**Goal:** Attach soil property values to each sample point.

Six soil property layers were sampled from OpenLandMap datasets (global coverage, 250 m resolution, lab-validated):

| Column              | Source Layer                           | Raw Units          | Conversion |
|---------------------|----------------------------------------|--------------------|------------|
| ph_0_30cm           | SOL_PH-H2O_USDA-4C1A2A_M/v02           | pH × 10            | ÷ 10       |
| organic_carbon_g_kg | SOL_ORGANIC-CARBON_USDA-6A1C_M/v02     | g/kg × 5           | ÷ 5        |
| sand_pct            | SOL_SAND-WFRACTION_USDA-3A1A1A_M/v02   | %                  | —          |
| clay_pct            | SOL_CLAY-WFRACTION_USDA-3A1A1A_M/v02   | %                  | —          |
| silt_pct            | Derived: 100 – sand – clay             | %                  | —          |
| bulk_density_g_cm3  | SOL_BULKDENS-FINEEARTH_USDA-4A1H_M/v02 | cg/cm <sup>3</sup> | ÷ 100      |

All layers represent the **0–30 cm depth** (topsoil), which is the agronomically relevant root zone for corn.

## Step 4 — Climate Metric Extraction

**Goal:** Capture seasonal weather variation that affects crop performance.

PRISM AN81m monthly climate data was used to derive six seasonal metrics per point per year:

| Column           | Months  | Variable                             |
|------------------|---------|--------------------------------------|
| annual_precip_mm | Jan–Dec | Total precipitation                  |
| summer_precip_mm | Jun–Aug | Total precipitation                  |
| spring_precip_mm | Mar–May | Total precipitation                  |
| mean_temp_c      | Jan–Dec | Mean temperature                     |
| summer_temp_c    | Jun–Aug | Mean temperature                     |
| vpd_max_kpa      | Jan–Dec | Mean maximum vapour pressure deficit |

Climate metrics were extracted at PRISM’s native 4 km resolution using `ee.Image.sampleRegions()`.

## Step 5 — County Assignment

**Goal:** Tag each GPS point with its Iowa county for yield data joining.

Each sample point was spatially joined to the **TIGER/2016 Iowa county boundaries** using `ee.Join.saveFirst()` with an intersects condition. This produced `county_name` and `county_fips` fields used in Step 6.

## Step 6 — USDA NASS Corn Yield

**Goal:** Attach a ground-truth corn productivity signal to each sample.

County-level corn yield data (bushels/acre) was fetched for all four years from the **USDA NASS QuickStats API** using the following query parameters:

- `commodity_desc:` CORN
- `statisticcat_desc:` YIELD

- `unit_desc`: BU / ACRE
- `agg_level_desc`: COUNTY
- `state_alpha`: IA

This returned county × year yield records, which were joined to sample points on (`county_name`, `year`).

## Step 7 — Quality Label Assignment

**Goal:** Convert continuous yield into a discrete three-class label.

Thresholds were selected based on Iowa’s historical USDA NASS averages (2010–2023), where the state mean is approximately 180 bu/ac:

| Label         | Threshold     | Meaning                                        |
|---------------|---------------|------------------------------------------------|
| <b>High</b>   | ≥ 200 bu/ac   | Exceptional yield year/location                |
| <b>Medium</b> | 160–199 bu/ac | Average performance                            |
| <b>Low</b>    | < 160 bu/ac   | Below-average, poor soil or climate conditions |

The yield signal is a **county-level proxy**, intentionally so: it captures the combined effect of soil quality, local climate, and agronomic suitability — which is the prediction target.

## Image Processing

After GEE exports completed, GeoTIFFs were converted to final JPGs using `convert_images.py`:

1. Read R, G, B bands from the GeoTIFF
2. Apply **2nd–98th percentile stretch** to normalise brightness across lighting conditions and seasons
3. Clip to [0, 255] and cast to uint8
4. Resize to **224 × 224** using LANCZOS resampling
5. Save as JPEG (quality=95)

The percentile stretch handles the wide variation in aerial imagery brightness caused by season, sun angle, and sensor calibration differences across NAIP collection years.

## Dataset Schema

Each row in `labels.csv` corresponds to one image:

| Column              | Type   | Description                       |
|---------------------|--------|-----------------------------------|
| filename            | string | Links to images/<filename>.jpg    |
| lat, lon            | float  | Centre coordinate of image patch  |
| county_fips         | string | 5-digit FIPS code (e.g., 19139)   |
| county_name         | string | Iowa county name                  |
| year                | int    | Crop year (2019–2022)             |
| naip_year           | int    | Year NAIP imagery was captured    |
| ph_0_30cm           | float  | Soil pH, 0–30 cm                  |
| organic_carbon_g_kg | float  | Organic carbon (g/kg)             |
| sand_pct            | float  | Sand fraction (%)                 |
| silt_pct            | float  | Silt fraction (%)                 |
| clay_pct            | float  | Clay fraction (%)                 |
| bulk_density_g_cm3  | float  | Bulk density (g/cm <sup>3</sup> ) |
| annual_precip_mm    | float  | Annual precipitation (mm)         |
| summer_precip_mm    | float  | Jun–Aug precipitation (mm)        |
| spring_precip_mm    | float  | Mar–May precipitation (mm)        |
| mean_temp_c         | float  | Mean annual temperature (°C)      |
| summer_temp_c       | float  | Mean Jun–Aug temperature (°C)     |
| vpd_max_kpa         | float  | Mean max VPD (kPa)                |
| corn_yield_bu_ac    | float  | USDA NASS county yield (bu/ac)    |
| quality_label       | string | High / Medium / Low               |

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## Dataset Statistics

- **Total samples:** 1,121
- **Years covered:** 2019, 2020, 2021, 2022
- **Iowa counties covered:** up to 99
- **Image size:** 224 × 224 px JPEG
- **Tabular features:** 14 numeric columns + 1 label

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## Validation

The assembled dataset was validated using `validate_dataset.py`, which checks:

- Missing columns or null values

- Out-of-range soil/climate values (physical plausibility bounds)
  - Image file presence on disk
  - Quality label class balance
  - Spatial coverage across Iowa counties
  - Year distribution
  - Pearson correlation of each soil/climate metric with corn yield
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## Reproducibility

The full pipeline is reproducible given:

1. A Google Earth Engine project (free academic access)
2. A USDA NASS QuickStats API key (free, instant)
3. Python environment: earthengine-api, geemap, requests, pandas, numpy, tqdm, Pillow, rasterio

Sampling is seeded per year (`seed=year`) ensuring the same GPS points are drawn on re-runs. GEE dataset versions are pinned by asset path.

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## Limitations

- **County-level yield** is a spatial proxy, not a field-level measurement. Intra-county variability is not captured.
- **NAIP capture year** may differ from the crop year by up to two years in counties with less frequent coverage.
- **OpenLandMap soil data** is at 250 m resolution — finer spatial variation within a 50 m patch is not represented.
- **PRISM climate data** is at ~4 km resolution and does not capture micro-climate effects.